RAG Lab Report: From Retrieval to Agentic RAG

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Models: all-MiniLM-L6-v2 (embeddings) and Qwen2.5-1.5B-Instruct (generation), run on a Colab T4.

Task 1: Chunking

(a) With size=100 and overlap=20 the handbook (718 words) gives **9 chunks**.

(b) Chunk 0 ends mid-sentence at "…a student cannot be excused", so the limit "for more than three weeks of lectures" is cut off. Chunk 1 starts mid-sentence at "seven days of returning to class", so the start of that rule (medical leave must be submitted within…) is missing. An embedding of a half-sentence can be misleading, and a generator may quote an incomplete rule. The 20-word overlap repeats the end of one chunk at the start of the next, so each sentence is complete in at least one chunk.

Task 2: Embed and Retrieve

(a) For "How many books can I borrow from the library?" the top chunk is chunk 5 with score 0.530. Chunk 4 is second with 0.438 and chunk 3 third with 0.157. The sentence that actually answers the question ("borrow up to three books") is in **chunk 4**, not chunk 5. Chunk 5 starts at "a loan period of fourteen days" and covers fines and reference books, so it is topically closer to "library" but does not contain the number. Chunk 4 mixes the end of the fee-refund section with the start of the library section, which dilutes its embedding. This shows that the top-1 chunk is not always the one holding the answer, and why top-k with k greater than 1 matters.

(b) The handbook never mentions chargers, so nothing in it truly answers this question. Dense retrieval matches meaning, not exact words, so it always returns the nearest chunk by embedding similarity, even when nothing in the handbook mentions chargers. It therefore returns the closest examination-related chunks, but with noticeably lower scores than a true match. The retriever has no way to say "nothing relevant" unless we add a score threshold.

Task 3: Recall@k

Chunk size (words)	Number of chunks	Recall@1	Recall@3
40	23	1.00	1.00
100	9	1.00	1.00
250	4	1.00	1.00

(a) All three sizes tie at Recall@3 = 1.00, so recall cannot separate them. The largest size is not automatically the best choice. With 250-word chunks the top-3 is about 750 words, which is the whole 718-word handbook. Recall@3 is then trivially perfect, but the prompt is large and full of irrelevant text, which costs time and can distract a small model. With 40-word chunks the same top-3 is only about 120 words. Size 100 is a reasonable middle: complete answers per chunk and a compact prompt.

(b) No question failed at any chunk size. The handbook is tiny (4 to 23 chunks) and each test question shares many words with its answer sentence, so this test set is too easy to expose failures. A harder test would use paraphrased questions and a longer document.

Task 4: Grounded Answer with Citations

(a) For "Can I take a reference book home?" the model answered "No. [1]". The handbook says reference books, theses and journals cannot be taken out, so "No" is correct. That sentence is only in chunk 5, so the citation [1] is correct when it points to that chunk.

(b) For the hostel-fee question the model replied "I could not find this in the documents." It did not invent a fee. The hostel section has no fee information, so this is correct.

(c) Removing that sentence takes away the model's instruction for what to do when the context lacks the answer. Without an explicit refusal instruction, a small model tends to fill the gap from its own knowledge or give a vague generic answer. The refusal behaviour comes from the prompt, not from retrieval, so prompt design is part of making RAG trustworthy.

Task 5: Hybrid Search with RRF

(a) For the query "F-4015": dense returned chunks [4, 6, 3], BM25 returned [4, 0, 2], and hybrid returned [4, 3, 2]. The chunk containing F-4015 is chunk 4, and **all three methods** placed it in their top 3 (rank 1 in each).

BM25's ranks 2 and 3 (chunks 0 and 2) have a score of 0.0. They are arbitrary ties, not real matches, yet RRF still gives them points. That is why chunk 2 appears in the hybrid result. Dropping zero-score BM25 hits would remove this noise.

(b) Dense Recall@3 was 1.00 and hybrid Recall@3 was 1.00, so hybrid did not help on the six test questions. With only 9 chunks, a top-3 already covers a third of the document and dense search finds the right chunk easily. On a real collection with thousands of chunks, product codes, form numbers, names and rare terms are poorly captured by embeddings, while BM25 matches them exactly. Fusion then recovers items one method would miss.

Task 6: HyDE

(a) The hypothetical passage written by the LLM was: "If you decide to withdraw from your studies before completing them, it is important to note that most universities have specific policies regarding refunds or cancellations. Typically, you will receive a refund based on the percentage of time you have already completed your course work and any remaining fees. It's advisable to check with your academic advisor or the financial aid office…" It is generic and contains no handbook facts. Its claim that the refund depends on "the percentage of time you have already completed" is wrong for this handbook, which gives 100%, 75%, 50% and 0% depending on the week of withdrawal.

(b) The correct passage (chunk 3, containing "refunded in full") was at **rank 2 with plain search and rank 2 with HyDE**, so there was no difference. The small model does not know this university's policy, so it could only produce a generic passage. The question's key words (money back, drop out) are also close enough to "refund" and "withdraw" that plain dense search already works on a 9-chunk document.

HyDE helps most when the question's vocabulary is far from the document's and the LLM knows the domain well enough to write realistic wording.

Task 7: Lost in the Middle

(a)

Question	First	Middle	Last
Library fine per day	True	True	True
Medical leave submission days	True	True	True
Courses allowed on probation	True	True	True

(b) No. All nine runs were correct, so the "lost in the middle" pattern did not appear. With three questions, one run each and 41 short passages (about 700 words), I cannot conclude anything, and the result may simply be a ceiling effect: the context is short and each question shares many words with its answer sentence. To make it more reliable I would use many more questions (50 or more), many more and longer distractor passages, confusable distractors, more positions, several shuffles per question, and a keyword check that also accepts digits ("7" as well as "seven").

Task 8: Agentic RAG with Support Check

Agent output:

```
step 0: SEARCH: Withdrawal policy details
step 1: SEARCH:
step 2: SEARCH:
step 3: ANSWER: If you withdraw in the first week, you receive a 75% refund of your
tuition fees. Use the "Withdrawal Form" to initiate the process.
```

Support check: both sentences were marked SUPPORTED.

(a) It made 3 SEARCH replies, but only the first had a query. Steps 1 and 2 were "SEARCH:" with nothing after it. The first query, "Withdrawal policy details", is sensible but vague. It merged both parts of the question instead of also searching for the form (for example "refund form"). The empty queries are probably the model writing its query on the next line, which the one-line parsing in `agent()` discards, so two of the four steps were wasted.

(b) Both sentences passed, but the checker was wrong on the second. The handbook says the form is **F-4015**, not a "Withdrawal Form". The model made up a form name, probably because the chunk with F-4015 (chunk 4) was never retrieved. The first sentence (75% in the first week) is correct. A 1.5B model used as a judge is too lenient: it saw the word "form" in the context and said yes. The answer also had no [n] citations, even though the prompt asked for them.

(c) Yes, it still searched. Output: `SEARCH: What is 2+2?`, then two empty `SEARCH:` lines, then `ANSWER: 2+2 equals 4.` The answer is correct but has no citation, and three of the four steps were wasted searches. For arithmetic this is harmless. It does show that the small model is not really deciding when retrieval is needed: it follows the "search first" instruction in the prompt. The repeated empty `SEARCH:` lines, the same pattern as in the handbook question, also show how fragile the home-made format is.

(d) Problems with the home-made SEARCH:/ANSWER: format: (1) the model can write an empty or mis-formatted line, as in steps 1 and 2; (2) parsing by string prefix and first line is fragile, and a multi-line answer is truncated; (3) the format can be ignored, and nothing validates it; (4) there is no schema for arguments, so there is no way to require a non-empty query; (5) it supports only one action and cannot easily ask for several searches at once. Function calling gives a declared tool name and typed, validated arguments, machine-readable output, and a standard place to return tool results.

Summary

On this tiny handbook, dense retrieval, hybrid retrieval and HyDE all achieved perfect or near-identical recall, so the main lessons came from failure cases: a top-1 chunk that did not hold the answer (Task 2), a generic HyDE passage (Task 6), and an agent that invented a form name while a weak checker approved it (Task 8). Bigger documents and harder questions would be needed to separate the methods.